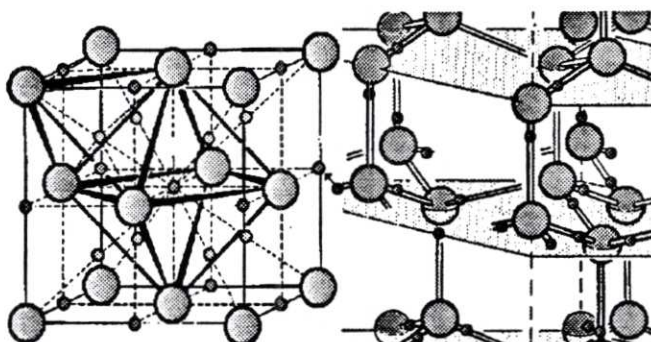


TO COLDLY GO WHERE NO ONE HAS GONE BEFORE

Metallic palladium (at left) is shown fully loaded with deuterium, an isotope of hydrogen. Heavy water (on the right) is the source of the isotopic fuel for "cold fusion".



COLD FUSION TIMES

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THE JOURNAL
DEVOTED TO
RELEVANT RESEARCH
and DEVELOPMENTS
OF SAFE, RELIABLE
POWER SOURCES

Columbo, Sri Lanka

2001: THE COMING AGE OF HYDROGEN POWER

by Arthur C Clarke, C.B.E.

Fellow of King's College, London Chancellor,
International Space university Chancellor,
University of Moratuwa

The following are brief excerpts from Prof. Clarke's Address to Pacific Area Senior Officer Logistics Seminar (PASOLS) 29 March 1993, Colombo, Sri Lanka.

"I'm very happy to be here today, even though I should really be in Washington ... in the uptown Theatre to celebrate the 25th anniversary - I can't believe it! - of 2001: A Space Odyssey.

Now, that movie provides a very good example of how difficult it is to predict the future. You may recall that in the film we showed the Bell System and PANAM - well, they've both gone, long before 2001. But I'm

Continued on page 3

Washington, D.C.

Cold Fusion and Alternative Fusion Are the High Points of U.S. Congressional Fusion Energy Hearing

by Eugene F. Mallove, Sc.D.

The Subcommittee on Energy of the House. Space, Science, and Technology Committee met on May 5, 1993 to discuss the status and funding of fusion energy. The hearings began at 1:30 pm in the Rayburn House Office Building and lasted to 6:00 p.m. with only a few 15-minute breaks for votes on the House floor.

Continued on page 2

In March 1989 electrochemically induced nuclear fusion reactions were reported, but were initially very difficult to reproduce. By mid 1993 many experiments have demonstrated that anomalous nuclear processes occur in palladium which has been highly loaded with deuterium.

COLD FUSION TIMES is devoted to novel research and developments in the field popularly known as "cold fusion", which has expanded considerably since 1989. This newsletter reports to its readers on the current state-of-the-art from around the world.

Cambridge, Massachusetts

Dr. Edmund Storms' (Los Alamos National Laboratory) Lecture at MIT

by M. Swartz, Cold Fusion Times

Dr. Edmund Storms' May 6, 1993 lecture at MIT, sponsored by the Department of Electrical Engineering and Computer Science, demonstrated further support to the fact that CF, and the heat generated, are real. He examined the "skeptics" most frequent "reasons" to dismiss cold fusion, and explained the *raison d'etre* for the confusion behind each "reason". He suggested calling the reactions CANR (for Chemical Assisted Nuclear Reactions).

After reviewing the recent experimental work in cold fusion he presented and discussed his own results. These results include the further defining the two types of palladium samples. Some work. Some do not work. Although it is well-known that many palladium samples simply do not work, wisdom is knowing which is which.

The ability to distinguish palladium samples which properly function depends in part

Continued on page 2

"... it was impossible to believe, back in 1945, that (satellite) TV relay stations could operate without a staff of engineers changing tubes and checking circuits"

It's not generally realised that the energy cost of going to the Moon is less than a hundred dollars in terms of kilowatt hours of electricity."

"It is now beyond serious dispute that anomalous amounts of energy are being produced from hydrogen by some unknown reaction."

Prof. Arthur C. Clarke

TABLE OF CONTENTS
ON PAGE 2

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CONTINUED FROM PAGE 1:

U.S. Congressional Fusion Energy Hearing

The hot fusion program was the focus up to about 4:00 pm, with the hot fusion ranks again coming to ask for further hundreds of millions to continue their work. After that, the heretofore outcasts -- cold fusion and aneutronic hot fusion -- were the subjects. So for the first time since the House Science, Space, and Technology hearing of April 1989, cold fusion really received an open airing before an important congressional committee. After the very positive reception at this meeting, it appears to this observer likely that further Congressional exploration of cold fusion research will occur. The "ice has been broken," with Congress showing that it is can no longer be intimidated by the foolish mockery of cold fusion by the scientific establishment.

Testifying for cold fusion was Dr. Edmund Storms of Los Alamos National Laboratory. Adding to the cold fusion case -- low-temperature energy extraction from water -- was Dr. Randell Mills of Hydrocatalysis Power Corporation, who presented his outlook and revealed new experimental achievements. During this "Alternative Programs" session, Dr. Bogdan Maglich of Advanced Physics Corporation presented a strong defense of his helium-3-deuterium collider scheme and attacked DOE's tokamak program. Dr. Klaus Berkner of Lawrence Berkeley Laboratory, who spoke about heavy ion inertial fusion during this "Alternative Programs" session must have felt very out of place on this panel of heretics; by the standards of the mainline hot fusion program he was a heretic too.

The hot fusion part of the meeting was chaired by Congresswoman Marilyn Lloyd (D- Tennessee), Congressman Dick Swett (D- New Hampshire) chaired the "Alternative Programs" session.

Panel One -- Opening Comments:
Dr. N. Anne Davies, Associate Director for Fusion Energy, DOE
Dr. Harold Forsen, Senior VP for Research and Development, Bechtel Corporation, Washington, D.C.

Panel Two -- TOKAMAK Programs
Dr. Paul-Henri Rebut, Director, ITER (International Thermonuclear Experimental Reactor), San Diego Co-Center, LaJolla, CA
Dr. Charles Baker, ITER, US Home Team Leader, Oak Ridge National Laboratory
Dr. Ronald C. Davidson, Director, Princeton Plasma Physics Laboratory

Panel Three -- Alternative Programs
Dr. Klaus Berkner, Associate Laboratory Director, Operations, Lawrence Berkeley Laboratory, Berkeley, CA
Dr. Bogdan C. Maglich, Chief Scientist, Advanced Physics Corporation, Irvine, CA
Dr. Edmund Storms, Staff Member, Los Alamos National Laboratory
Dr. Randell L. Mills, Hydrocatalysis Power Corporation, Lancaster, PA

Defining the spirit of the session, Congresswoman Marilyn Lloyd said in her opening remarks, "...our fusion program may be entering a state of reassessment. This, in part, may be the result of not having produced a fusion energy plant. Or maybe it's because we are all becoming more aware of the pressure on federal research funds."

Dr. N. Anne Davies's prepared statement acknowledged the magnetic fusion program's objective of a "Demonstration Power Plant" by about 2025 and an "operating commercial power plant" by about 2040. Notice that the hot fusion people never say just "by 2025" or "by 2040." It's always "by about...20XX" The FY 1994 budget request, Davies said was

Continued on page 4

TABLE OF CONTENTS

ARTICLE	AUTHOR	LOC'N
2001 - The Coming Age of Hydrogen Power	Clarke	1
Cold Fusion Impact	Fox	2
Cold Fusion and History	Rothwell	6
Miles' Letter to J. Phys. Chem.	Miles	4
Storms' Letter to Congress	Storms	4
Storms' MIT Lecture	Swartz	1
Taubes Review	Mallove	7
U.S. Congressional Hearing	Mallove	1
What's Happenin	Various	11

CONTINUED FROM PAGE 1:

Dr. Storms' MIT Lecture

upon noting that the good electrodes do ****NOT**** increase in volume as much inactive pieces upon loading. Dr. Storms' data supports this criterion. As discussed in his FUSION TECHNOLOGY article (3/93, Volume 23, p230): "Measurement of Excess Heat from a Pons-Fleischmann-type Electrolytic cell using Palladium Sheet":

"The first [active] sheet contained 0.8% excess volume after having been deloaded from its maximum deuterium/palladium (D/Pd) ratio of 0.82 to 0.73, and the second [inactive] sheet contained 13.5% excess volume while at its maximum ratio of 0.75."

BRIEF SUMMARY: The excess heats were 20% (of input power) with a peak of 7.5 Watts excess. The equipment included sealed calorimeters, stirred, with a spatial distribution of temperature sensors. The distributed cluster of temperature probes confirmed that the heat output was from the cathode.

This lecture completed another year of CF seminars arranged by Profs. Smullin and Hagelstein. Like the others, this seminar was also well-attended, but this time, no negative comments were made. With the relative absence of hot fusioners, the distribution of scientists seemed displaced towards metallurgists, material, environmental, energy, and CF scientists.

COLD FUSION IMPACT

by Hal Fox

[The following is an excerpt from this excellent book which goes into considerable detail along many important issues]

III. IMPACT ON GOVERNMENTS

Governments around the world are beginning to sense the revolutionary impact of cold fusion energy. To predict future patterns, it is valuable to consider why strongly different responses to cold fusion have already emerged in various nations.

GLOBAL RESPONSE: A STUDY IN CONTRAST

Immediately after the announcement of the important discovery of cold fusion by Fleischmann and Pons, scientists worldwide began working on replication of the Fleischmann-Pons Effect (FPE). By May 10, 1989, a lengthy article in the India Times. (Bombay) reported on the successes of several teams of Indian scientists. By July 31, 1989, Japanese successes led to a cold fusion workshop in Japan, and the next day, 80 scientists from 15 universities were assigned to work on cold fusion. Reportedly, three national coordinating groups in Japan were formed to coordinate experimental, theoretical, and applicational projects. By August 1989, India had 18 teams of four research workers in at least three national research laboratories working on cold fusion, with one research team working on enhancing the production of neutrons for medical and scientific purposes. In mainland China, a special conference on cold fusion was held in late summer of 1990. Over 100 papers were presented, nearly all of them positive. A visiting Chinese scientist traveled to Japan where he learned there were over 90 groups in Japan's colleges and universities working on cold fusion. In the U.S., this scientist reported On the strong enthusiasm for cold fusion developments which he found in China and Japan. He could not understand why there was so little support for cold fusion in the United States.

In America, the fusion research teams were nearly all working on hot fusion and lacked the experimental knowledge of working with electrochemical fusion. As a group, the hot fusioners had effective control over technical inputs to those in control of DOE funding. By giving wide publicity to failures to replicate cold fusion, the hot fusioners effectively prevented higher officials in the U. S. government from receiving the truth about the gradual and increasing successes of cold fusion research. Even though many of these successes came from DOE-funded national energy laboratories, the pressure of dollar allocation in favor of existing hot fusion research resulted in the cutoff of funding to most of the cold fusion research in U. S. government-supported national energy laboratories. Of these ten laboratories only Los Alamos National Laboratory had any significant cold fusion work that was being conducted at the end of 1992. The end result has been that the Bush administration had been made the unwitting party to a disgraceful treatment of an important new energy science. As the truth becomes known, it is likely that a Congressional investigation will be launched, possibly with vigorous vocal condemnation of the Bush administration and Secretary of Energy W. Atkins.

Continued on page 5

Because of rapidly developing technologies in cold fusion research, COLD FUSION TIMES is available for home or office delivery. COLD FUSION TIMES brings up-to-date, relevant information to its readers, and an open forum for all views on cold fusion. The COLD FUSION TIMES is not responsible for the comments of its reporters or readers whose words remain protected by the First Amendment of the U.S. Constitution. Questions, comments, suggestions, and contributors are welcome.

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COLD FUSION TIMES

CONTINUED FROM PAGE 1:
**2001: THE COMING AGE OF
 HYDROGEN POWER**

happy to see that the Hilton, which we also showed in 2001, is still here, though not yet in orbit! This proves how impossible it is to predict social and political developments.

....

However we can, to some extent, anticipate technological developments, by observing what's going on in science and engineering. But the problem there is predicting when things will happen, even though one can be quite certain that they will.

A good example is provided by my 1945 paper on communications satellites, which I imagined would be large, manned spacestations. When I wrote that, World War II was still in progress, and I was working on Ground Controlled Approach Radar, which had the then enormous number of something like a thousand vacuum tubes in it, at least one of which would blow everyday. So it was impossible to believe, back in 1945, that TV relay stations could operate without a staff of engineers changing tubes and checking circuits. But of course, the transistor and the solid state revolution came along within a few years, and what I'd assumed would have to be done by large manned stations could be achieved by satellites the size of oil drums. So everything I imagined would be done around the end of the century happened decades in advance.

...

So now I want to change the subject completely, to something perhaps even more important than the communications revolution. But first I'd like to mention a bit of forgotten history. In December 1903, Orville & Wilbur staggered off the ground in North Carolina, and made the first controlled flight in a heavier-than-air machine.

...

Yet for five years, Washington didn't believe that the Wright brothers had actually flown - because everybody knew it was impossible: leading scientists were still writing papers proving it couldn't be done. Not until the Wrights went to France and started giving public demonstrations did the boys in the War Department say "My goodness, these things really can fly. Perhaps they may even be useful for reconnaissance. We'd better look into it". And they did - five years late. Well, history has just repeated itself, with what's been (perhaps inaccurately) named 'Cold Fusion'.

You all know of course, that the Sun is powered by the fusion of Hydrogen atoms, when they combine to make helium. Tremendous efforts have been made to reproduce this reaction on earth and produce virtually unlimited amounts of energy.

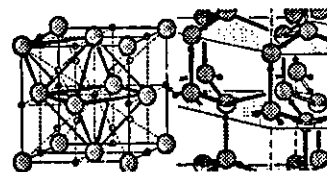
...

Literally billions of dollars have been spent in efforts to reach the multi-million degree temperatures in the heart of the Sun, where this reaction occurs. One day these experiments will succeed, but so far only a few percent of the input energy has been obtained, for very short periods of time.

However, just four years ago, two scientists named Pons and Fleischmann claimed to have achieved 'Cold Fusion' at room temperature in certain metals saturated with deuterium, the heavy isotope of hydrogen.

...

This, of course, created a world-wide sensation, and many laboratories tried to repeat the experiments. They all failed, and Pons



Dr. Arthur C. Clarke in Sri Lanka with his favorite computer and (?) parasentient mammal - Photograph by The Washington Post (3/9/92) by Robert Nickelsberg

"It is now beyond serious dispute that anomalous amounts of energy are being produced from hydrogen by some unknown reaction."

and Fleischmann were laughed out of court. That was the last anyone heard of them for a couple of years.

But meanwhile, there had been an underground movement of scientists who believed that there might be something in all this business, and started experiments of their own - often in defiance of their employers. Pons and Fleischmann went to France just like the Wright Brothers! - and are now working in a laboratory near Nice, financed by a Japanese consortium, Technova.

....

It is now beyond serious dispute that anomalous amounts of energy are being produced from hydrogen by some unknown reaction.

The term 'cold fusion' - 'C/F' - has stuck because no one can think of anything better. However, the sceptics who originally pooh-pooed the whole thing (appeared to have) had a very good point. If it really was fusion, (they claimed) the experimenters should be dead! Where were the neutrons and gamma rays and tritium and helium - the lethal 'ashes' such a reaction should produce? Well, they have now been detected - but in quantities far too small to account for the energy liberated. The theoretical basis of C/F is therefore still a major mystery - as was the energy produced by radio-activity and uranium fission when they were first discovered.

...

Twenty years ago, when OPEC quadrupled oil prices, I remarked "The age of cheap power is over - the age of free power is still fifty years ahead." I may have been slightly too pessimistic.

...

Now please fasten your seat-belts: after these modest daydreams, I want to really stretch your imaginations. ... In 1982, I published 2010: Odyssey II and dedicated it to my friend Cosmonaut Alexei Leonov - and to Academician Andrei Sakharov, ... and in the novel I suggested that, in his enforced solitude, he'd invented a spaceship engine based on these principles.

...

He didn't, of course, so that's a piece of fictitious history. However, three Russian scientists who have indeed been working on nuclear propulsion for rockets have now got into the cold fusion act, and they have just published some startling results in Physics Letters A, one of the world's leading scientific journals. They are obtaining about five times their energy input in gas mixtures, not solids, and at temperatures of up to 1800 C. Now this is not exactly 'cold' fusion - but it's certainly ice-cold compared with the tens of millions of degrees the hot fusioners are talking about.

And it's very interesting indeed from the point of view of rocket propulsion. If a plasma fusion rocket could be developed, it would open up the solar system, just as the airplane opened up this planet. It's not generally realised that the energy cost of going to the Moon is less than a hundred dollars in terms of kilowatt hours of electricity. The fact that the Apollo round tickets cost about two billion dollars per passenger is a measure of the chemically-fuelled rocket's inefficiency.

Continued on page 8

CONTINUED FROM PAGE 1: U.S. Congressional Hearing

\$347.6 million, supporting four program elements: conducting D-T experiments at Princeton in the TFTR, participating in the engineering design phase of ITER, initiating the detailed design of the TPX (another proposed machine for Princeton), and continuing a base program of physics and technology support.

Dr. Davies's prepared statement, loaded with ponderous bureaucratese, says at one point: "The Operating Expenses request also provides \$9.2 million in Program Direction funds for the salaries, benefits, travel and other expenses associated with 81 full-time equivalents required to administer the Fusion Energy program by the Headquarters staff and those at DOE Field Offices; and \$4.9 million in the Planning and Projects subprogram to support the program's legal obligation to the Small Business Innovation Research Program." Just a fraction of that administrative overhead would fund a much needed cold fusion research effort in universities and small businesses. Another farce of the current DOE fusion program: the published SBIR requests as they now stand are written up to explicitly exclude cold fusion proposals.

The other hot fusioners who followed Davies reiterated what their leader had said, but there were some revealing gems — explicit warnings NOT to be diverted by other non-tokamak hot fusion concepts. An example was a part of the testimony of Dr. Harold Forsen of Bechtel Corporation: "I worry very much about not getting on with the D-T fueled tokamaks at the expense of too much second guessing of what might be cheaper, smaller or fueled with more exotic, remote fuels." The U.S. hot fusioners plan to spend an "estimated" \$450 million between 1992 and 1998 on "Engineering Design Activities," which are then to be followed "if the [international] parties agree," by a seven-year construction phase so that ITER could begin to operate in 2005.

Congressman Harris Fawell of Illinois diplomatically showed his impatience with the hot fusion program. He read the following statement into the record: "Madam Chairman, I share your view that the fusion program may be entering a state of reassessment. And, in fact, this reassessment may be long overdue. I was particularly struck by the recent remarks of Dr. Robert L. Hirsch at the March 5, 1993 meeting of DOE's Fusion Energy Advisory Committee. Dr. Hirsch, who was once head of DOE's magnetic fusion energy program and is now a Vice President of the Electric Power Research Institute (EPRI) made the following observations:

- *DT tokamak and laser-fusion reactors as currently envisioned will be extremely complex, highly radioactive, likely to be highly regulated, and costly.
- * Even if DT or laser fusion reactors had the same capital costs as a fission reactor — an enormous challenge — fusion reactors would lose out to advanced fission reactors, which are a reliable, known quantity.
- * None of the very few fusion-knowledgeable utility people he had spoken with believes that tokamak or laser fusion reactors, as currently envisioned, would be acceptable to the electric utilities.
- * There are some enormous materials problems related to DT fusion. There are no qualified materials today for DT fusion reactors. In the absence of development of a low activity material — a very costly and time consuming undertaking — you will have to effectively rebuild your fusion reactor every 5-10 years and dispose of many times the amount of radioactivity that would come from a fission reactor of the same power level.
- * And then there's ITER. If tokamak reactors, as currently envisioned, aren't acceptable, can ITER possibly be justified? If you build ITER, it will become the flagship of fusion and will likely eliminate the chance of serious funding for alternate concepts.
- If what ITER represents is seriously considered in public debate, there is a high probability that ITER will not be supported and the fusion program could collapse.

Dr. Hirsch closed his remarks with several recommendations including:

- * Scale-up of alternate R&D concepts as fast as possible.
- * Don't stop tokamak or laser fusion, but cut them back and reorient them in more acceptable directions.
- * Get off the DT fuel cycle to avoid frequent reactor reconstruction, large quantity radwaste disposal, and expensive materials development.

I am also concerned about other aspects of the planned fusion energy program, including:

- * The role of, and the need for, a new proposed tokamak device, TPX, for which the DOE budget documents provide neither a cost estimate, a time schedule, nor an understandable rationale of why it is so important.
- * The introduction of tritium into the TFTR later this year — not only will this be done in a heavily populated area, raising safety and other environmental concerns — but it will also require expensive decontamination and decommissioning of the machine. Is the science we will get worth this cost, or would it be more cost-effective to rely on JET, which has already used tritium?
- * Finally, the continued operation of other tokamaks, including Alcator C-Mod and the DIII-D tokamaks, in this budget climate means that we cannot afford to pursue promising alternate concepts. Are we really getting our money's worth from them?

Dr. Randall Mills of Hydrocatalysis Power Corporation, whose heat-producing experiments with ordinary water-nickel-potassium carbonate cells are increasingly highly regarded in the cold fusion field made his presentation. His colleague Robert M. Shaubach of Thermacore, Inc. accompanied Mills but did not make a presentation. Mills's opening remarks concisely summarized what the Lancaster, PA effort is all about: "Hydrocatalysis Power Corporation (HPC) has an extensive theoretical and experimental research program of producing energy from light-water electrolytic cells. HPC and Thermacore, Inc., Lancaster, PA are cooperating in developing a commercial product. (Thermacore is a well respected defense contractor and its expertise is in the field of heat transfer.) Presently, all of the demonstration cells of HPC and Thermacore produce excess power immediately and continuously. Cells producing 50 watts of excess power and greater have been in operation for more than one year. Some cells can produce 10 times more heat power than the total electrical power input to the cell. A steam-producing prototype cell has been successfully tested. The source of excess energy is NOT from fusion or other nuclear reactions. Energy is released in a catalytic process whereby the electron of the hydrogen atom is induced to undergo a transition to a lower electronic energy level than the 'ground state' as defined by the usual model of the atom. The lower energy states of hydrogen have been identified in nature. HPC and Thermacore are conducting other experiments to demonstrate that this lower-energy form of hydrogen is the 'ash' of heat producing cells. A summary of the technology follows. The [original] experiment has been scaled up by a factor of one-thousand, and the scaled-up heat cell results have been independently confirmed by Thermacore, Inc. Patents covering the compositions of matter, structures, and methods of the HydroCatalysis process have been filed by HPC worldwide with a priority date of April 21, 1989. HPC and Thermacore are presently fabricating a steam-producing demonstration cell."

Dr. Mills continued with a further elaboration of the benefits of HPC's technology, as well as a discussion of possible evidence for the existence of below ground state hydrogen atoms. At one point the statement says, "The Mills theory predicts exactly the spectral observations of the extreme ultraviolet background emission from interstellar matter (given by Eq.(1) with n1 and n2 being fractions), which characterize dark matter, and HPC is conducting further verification of data that demonstrates that the lower-energy form of hydrogen is the 'ash' of heat producing cells."

Of course, many cold fusion people who accept the Mills excess heat experimental claims do not accept Dr. Mills's theory and are inclined to believe that other forms of ash — nuclear ash — may be arising from the deuterium in ordinary water. There is also the question about whether sodium carbonate electrolyte also facilitates the heat reaction — Bush and Eagleton say it does, the Indians at BARC say it does. Mills, on the other hand, says that sodium carbonate electrolyte doesn't give rise to excess heat, a fact that he says supports his catalytic theory. (By his own assertions, Dr. Mills has, in effect, presented this sodium versus potassium issue as a potential Achilles heel of his theory.)

Dr. Edmund Storms of Los Alamos National Laboratory read his prepared statement into the record. His statement might be viewed as fairly "mainline" cold fusion thinking, with a few exotic twists that will likely further enrage the folks who think that high-energy nuclear physicists have a monopoly on wisdom about possible nuclear reactions.

Dr. Storms's statement: "Madam Chairman, I want to thank you for this opportunity to present new and important results about a phenomenon that has been conventionally called 'Cold Fusion.' Starting with the work reported in 1989 by Drs. Pons and Fleischmann, many observations have indicated that it is apparently possible to initiate nuclear reactions in certain metals near room temperature and that these reactions result in significant heat production as well as various nuclear products. I am not speaking for the Los Alamos National Laboratory on this subject because policy in this area has not yet been formulated. The Laboratory does not want me to pre-empt this process. Nevertheless, a careful and extensive examination of available information as well as personal research has convinced me and many other scientists that this phenomenon is real and, I believe, may have important consequences to the U.S. Much skepticism and frustration resulted from lack of reproducibility during early experiments. For this and other reasons, many scientists still believe that positive results are not possible. However, the phenomenon is now reproducible using a variety of techniques. Excess heat production has occasionally approached useful levels and the many positive results are widespread in a variety of peer reviewed scientific journals and conference proceedings. One might ask what consequences might result if only a fraction of these assertions is true? FIRST, the theory of electron structure in a metal and its interaction with the atomic nucleus needs to be reexamined. I would expect such a reexamination to produce even more useful insights. SECOND, experiments indicate "a strong potential" for creating energy that has no identified pollution or significant radioactivity, and is essentially inexhaustible. THIRD, some speculate there is a "possibility" of

Excerpts from Dr. Edmund Storms' (Los Alamos) Letter to Congressman Dick Swett and Congressional Hearing

... A careful and extensive examination of available information has convinced me and many other scientists that most of the original claims made by Drs. Pons and Fleischmann have been supported by many researchers world-wide. The phenomenon is now reproducible if correct procedures are used, the magnitude of excess heat production has occasionally approached useful levels, and the results are now described in many peer reviewed scientific journals. In addition, the phenomenon has been demonstrated under a variety of conditions including those employed by conventional physicists. However, a satisfactory explanation is still lacking. For this and other reasons, the reality of the phenomenon is not generally accepted. If this were just another example of conventional science being slow to accept new ideas, I would be content to wait for nature to take its course. However, this phenomenon is of such importance that none of us can afford to wait. The need for a cheap and CO₂-free energy source is becoming increasingly important. Cold Fusion has the potential for solving this problem while leading to a complete change in the world's energy economy, a new and more useful theory of nuclear interaction in a periodic lattice, and a less messy way to make tritium and other isotopes. If changes are not made in the approach now taken by the US, the Japanese will dominate this technology. These amazing claims can be better believed if some of the observations are understood.

A fusion reaction between two deuterium nuclei appears to occur in palladium and other metals when the deuterium/metal ratio exceeds a certain value. Because this high loading is difficult to achieve, heroic methods must be used. These have included the now classical electrochemical method, gas discharge in deuterium at low voltages, ion implantation of D(+) at high voltages, and diffusion assisted gas loading. Apparently, the method is not important as long as the necessary deuterium concentration is achieved. Different nuclear reactions appear to be initiated depending on the chemical environment and the flux of deuterium. These reactions have resulted in the production of tritium, helium, and various products that result from nuclear reactions between palladium and deuterium.

Heat production has reached levels that not only exceed by many orders of magnitude any conceivable chemical source, but occasionally exceed breakeven if used as an industrial source of energy. Skeptics who suggest that the heat has a chemical origin fail to note that this explanation would suggest a phenomenon that would be more amazing and useful than if the heat were caused by a nuclear reaction. Power density above 1000 watt/cm³ of palladium has been reported using thin films as well as bulk metal. This is higher than the power density of a conventional nuclear reactor. It is clear that if such high power density can be sustained, the rarity of palladium would not be a problem. Although most studies have made heat at temperatures between 20 deg and 100 deg C, a few examples exist of heat being made at temperatures above 450 deg C using electrolytic cells based on fused salts. The challenge is to make the process work at still higher temperatures. Surprisingly, significant heat has also resulted when normal water containing Li(2)CO(3), Na(2)CO(3), K(2)CO(3), or Rb(2)CO(3) is electrolyzed using high surface area nickel as the cathode. Some evidence has been reported showing that the heat is caused by proton pickup causing atomic transmutation to the next higher element. Significant heat and nuclear products have also been reported when palladium was bombarded by low energy deuterium ions produced in a gas discharge. This method, developed in Russia, shows convincing signs of producing sufficient heat to make scaleup for industrial use very practical. Clear evidence for nuclear products has also been reported. This technique appears to be easier to scaleup than is presently experienced using the hot fusion method.

In spite of the perceived limitations of the initial work reported by Drs. Pons and Fleischmann, the absence of adequate theory, or the general skepticism of the scientific profession, this field is moving ahead. While support is very limited in this country, the Japanese government through MITI and several companies are funding and encouraging a broadly based plan of research involving industry and university cooperation. Four teams of scientists from Japan are now visiting laboratories in the US, Italy, and India in search of ways to develop joint efforts. Drs. Pons and Fleischmann are now working in France under Japanese sponsorship. Significant studies are also occurring in India (Bhabha Research Centre), Italy (University of Rome, Laboratori Nazionali Di Frascati), Russia (Lutich), and China. Presently, the only regular source of funding in the US is provided by EPRI (Electric Power Research Institute) and to a lesser extent by Southern California Edison. Various other companies are slowly getting into the field. Studies are continuing at the Los Alamos National Laboratory by three groups with very modest and erratic support. Each of these groups has obtained significant positive results in different aspects of the phenomenon.

Work in the US is lagging because the Patent Office refuses to issue patents, and the DOE, as well as other agencies, refuse to acknowledge the importance of the field. I do not need to remind you that if the possibilities that this field offer are eventually realized, our failure now to create an intellectual infrastructure and legal protection will be very damaging. If changes are not made soon, this technology will be dominated in the future by others. Even those who suffer from cautious skepticism should see an advantage to promoting a field having such extraordinary potential.

Very sincerely,
Edmund Storms

Continued on page 8

COLD FUSION TIMES

CONTINUED FROM PAGE 2:
COLD FUSION IMPACT

Little or no electrochemical work was being performed in the ten U. S. tax-supported nuclear laboratories at the time of the FPE announcement. Secretary Watkins asked the national laboratories to investigate the FPE claims even though few scientists at these laboratories were skilled in the art. However, many Department of Energy funded scientists began to attempt the replication of the FPE. Most of the initial reported results were negative. Later, there were positive results from Brookhaven, Oak Ridge, and Los Alamos National Laboratories, where scientists were successful in replicating cold fusion. Subsequently, thanks to the continued efforts of many (200 - 300) dedicated scientists around the world, research work in cold fusion continued; professional meetings were held; papers were prepared, peer-reviewed, and published; and the news gradually spread that cold fusion was real but not fully understood nor easy to replicate.

On July 10, 1992, the powerful Japanese Ministry of International Trade and Industry (MITI) announced its full economic support for cold fusion and proposed to spend around 3 billion Yen (\$25 million of U. S. dollars) on a four-year project beginning in fiscal 1993. A Japanese scientist stated that there are no more cold fusion skeptics in Japan! This dramatic difference in early cold fusion achievements requires some explanation.

RESPONSE IN INDIA AND JAPAN

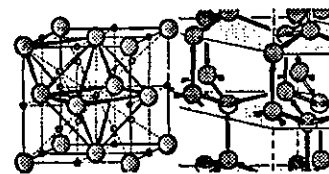
India is a land of 800 million people. The growth in energy production is a critical national priority. Therefore, a few government-supported scientists had for several years been experimenting with various forms of nuclear fusion to help solve the national energy problem. Some Indian scientists were working in the area of cold fusion and the electrolysis of heavy water, seeking the means by which the vast energy of water (often of the deuterium in water) could be unleashed.

When Pons and Fleischmann announced their incredible discovery, scientists in India had only to modify some of the experiments they were already performing. Several teams from nuclear energy laboratories in India replicated the Fleischmann-Pons Effect (FPE) within a few weeks. The exploitation of India during its colonial period left a nation with a large population, minimal industry (as compared to USA or Japan), a good educational system with a high English-language literacy (and thus inclined to read English-language journals), and a strong desire to be industrially independent. India's national energy policy is directed more to internal survival and raising the standard of living internally than to global marketing.

Japan, in contrast, emerged as the post World War II leader in manufacturing and marketing of high technology products. Japan now has one of the world's highest standards of living, a large positive balance of payments, and a strong national desire to achieve in the development of important new technologies. Japan's goals appear to be global and driven by wealth accumulation, while India's and China's goals appear to be centered on solving their internal development problems using their own human and natural resources.

These two dissimilar nations -- India and Japan -- have emerged as two of the best organized and most successful in developing early international leadership in this exciting new area of cold fusion. Both nations initiated early research work under national direction for the development and exploitation of cold fusion phenomena.

However, the continued negative comments about cold fusion from prestigious publications, such as *Nature* and *Scientific American*, caused government official support of cold fusion to diminish in India. Meanwhile, in Japan, the successes were not waned by the negative English-language press. Cold fusion research continued because the pay off had such an enormous potential and cold fusion became gradually accepted. India's cold fusion work continued mainly under the active leadership of Dr. M. Srinivasan of the Bhabha Atomic Research Centre at Trombay. Although there was decreasing official support, the BARC scientists have some choice in research projects. Therefore, many of these scientists continued to explore the new science of cold fusion. By setting an active example in energy development, India may yet strongly influence the national energy policy of many other developing nations. The Srinivasan encouragement of continued investigation of cold fusion, especially of the light-water electrochemical cells [2] may well be the catalyst for more extensive government-approved cold fusion research in India.



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IN REPLY REFERTO:
3910
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13 Apr 93

Dr. A. J. Nozik, Senior Editor
The Journal of Physical Chemistry
National Renewable Energy Laboratory
1617 Cole Boulevard
Golden, Colorado 80401

Dear Dr. Nozik:

In reply to your letter of 18 March 1993 rejecting my manuscript titled "Anomalous Effects Involving Excess Power, Radiation, and Helium Production During D_2O Electrolysis Using Palladium Cathodes", I would like to make some comments for the record regarding this controversial topic.

First, the letter I sent with my manuscript clearly requested the selection of competent and impartial referees. My suggested referees (Drs. E. M. Eyring, H. Gertscher and V. Rehn) were carefully chosen on this basis. However, your selection of referees such as Dr. Nathan Lewis (probably Reviewer 2) or Dr. Steven Jones (probably Reviewer 3) or similar referees with well-known polarized positions did not meet my request. Manuscripts reporting measurable excess power or helium production can be guaranteed a negative review by scientists such as Nathan Lewis, Steven Jones, David Williams, John Huijenga, Frank Close or Douglas Morrison regardless of scientific merit. The selection of such reviewers provides an easy pathway for avoiding the publication of my paper. It would require considerable backbone to publish a positive paper on this subject and then face the outrage of other scientists who have been out-touched with any new results in this field since 1989. I could readily accept the rejection of my manuscript if impartial referees had been selected such as Reviewer 1.

Second, I would like to address some comments made by Reviewer 2, especially his major argument that we "present no valid control experiments". We indeed ran six valid control experiments involving both Pd in D_2O and Pd in H_2O . This fact is discussed in my abstract as well as on page (7) of my manuscript and also in references (8) and (9). Helium was never detected in any control experiment. I challenge any critic using contamination arguments to explain how helium could be present in all eight experiments showing excess power and not be detected in any of the six control experiments. Due to the sporadic nature of the anomalous effects, it would be difficult to show that longer rods always yield proportionately larger amounts of heat and helium as suggested by Reviewer 2. The key relationship is between the amounts of excess power and helium observed and not the lengths of the palladium rods. Many experiments simply do not produce any anomalous effects regardless of the palladium cathode length.

The argument that Lewis et al., Albagli et al., Harwell, and Gai et al. did not detect any anomalous effects, as cited by Reviewer 2, proves very little other than the fact that these effects are elusive. I can also show results from many palladium- D_2O experiments involving months of work where no anomalous effects were detected. This includes my first publication on this topic (*J. Fusion Energy*, Vol. 9, pp. 333-336, 1990). To my knowledge, the scientists cited above have not performed any additional experiments on this topic since 1989.

The claims by Reviewer 2 that "calorimetry should be done in a closed system only" and "reports of neutral heat balance have been almost universally reported in such systems" is both

out-dated and false. This reviewer is obviously not following the field since he apparently has never heard of the calorimetric work of Dr. Michael McKubre of Stanford Research Institute or by Dr. Edmond Storms of Los Alamos National Laboratory that reports significant levels of excess heat in closed systems. The amounts of excess heat observed for these closed systems is in fact, in excellent agreement with my measurements (see reference 9). If experiments are performed properly, accurate calorimetric results can be obtained with open systems. However critical calorimetric errors such as those present in the N. Lewis work must be avoided (see reference 9).

Third, I would like to reply to some comments by Reviewer 3. It is true that the exposure of dental film does not provide energy information; nevertheless, the fogging of the film cannot be easily dismissed. After all, radioactivity itself was discovered in 1896 by Henri Becquerel due to the fogging of photographic plates. The rare observation of anomalously high counts by the use of a Geiger-Mueller detector during the same time period as the dental film exposure and excess power generation defies any simple explanation.

I found it interesting that all three reviewers ignored the fact that completely new experiments again showed simultaneous evidence for excess power, helium production, and anomalous radiation as stated in the abstract and discussed on page (8) of my manuscript. The effect of helium diffusion through the glass was eliminated by extrapolating measurements back to the time of the gas sample collection (see figure (2) of my manuscript). Furthermore, the helium measurements were much more accurate than in our previous experiments. These results again yielded 10^{11} - 10^{12} helium atoms per second per watt of excess power. It simply cannot be coincidental that cells producing more excess power always yield more helium and that the amount of this helium corresponds to what would be expected as a nuclear ash for typical fusion reactions.

Sincerely

Melvin H. Miles

DR. MELVIN H. MILES
Ph.D. Physical Chemistry
NAWCWPNS Fellow



Cold Fusion Impact
by Hal Fox, Ed. Fusion Facts

The April 13, '93 letter above by Dr. Melvin Miles demonstrates the frustration felt through out the cold fusion field within the United States. We salute Dr. Miles, and the rest of the women and men in the trenches deserve the deepest gratitude of all for their persistent courage and diligence.

Cold Fusion And History

by Jed Rothwell, Cold Fusion Research Advocates

Throughout history, most new ideas and theories were, at first, rejected out of hand. This is true of correct ideas, incorrect ones, big ones, and little ones, including many that do not seem controversial now. A century ago, when someone suggested that the newly invented "zipper" be used to close the fly on men's trousers, the idea was met with gales of laughter and derision. When I suggested in 1979 that the newly invented microprocessor might eventually take away minicomputer business, the management of my company (and of IBM, and DEC) thought the idea was absurd. The acceptance — or rejection — of an idea cannot be used to predict its future success.

Experimental evidence, no matter how simple and overwhelmingly convincing, is usually ignored for a period of years. You can fly an airplane in front of hundreds of people right next to the urban transport trolley line; you can circulate photographs, data, and affidavits from leading citizens saying that they have witnessed flights. It does not matter what you do; the leading establishment scientists, the Army, the Scientific American, and the New York Times will go on denying that airplanes exist year after year, and they will never bother to check the facts firsthand. In a world that has never seen a glass lightbulb, you can string electric incandescent lights on poles outside a laboratory. Ordinary people will come from miles around to see it, but the experts will say it cannot be true. They will denounce you, and they will not bother coming out to check, even if you are the most important inventor in the world, and you have backing from J.P. Morgan. You can show a video of a boiling cell, and hand out data, and be replicated in hundreds of the world's leading laboratories at levels of confidence as high as sigma 90, but the "skeptics" will ignore you, even if Toyota has given you carte blanche support.

These patterns of history have been repeated over and over again. They should not surprise anyone. It is disappointing that the Scientific American is as stupid today as it was in 1879, or in 1908, but is not surprising, because human nature has not changed. Experts are human beings, after all, they make dreadful mistakes as often as anyone, only on a grander scale. When a cook makes a dreadful mistake, someone dies of food poisoning; when a programmer fouls up, a computer crashes somewhere, and somebody loses a bundle of cash. When industrialists, experts, and leaders make mistakes, it leads to events like the Titanic Disaster, the Battle of the Somme, the bankruptcy of the Pennsylvania Railroad, the Challenger Explosion, the S&L collapse, and other horrors. The treatment of CF in the U.S. is merely the latest example of stupendous miscalculation and folly.

Two very well documented examples of this tendency exist: Both the Wright's airplane and Edison incandescent lightbulb met firestorms of establishment scientific opposition, which went on for years. For people interested in these examples, I recommend the Wright's "authorized" biography, The Wright Brothers by Fred C. Kelly (with help from Orville Wright; and a superb "warts and all" biography of Edison titled A Sreak of Luck, by Robert Conot.

Regarding Edison, let's look at the situation in late 1879. This was 3 years after Bell invented the telephone, and about six months after Edison got serious about the incandescent light. He had gone through four design iterations, open air, vacuum, partial vacuum... He and his crew were doing "teeth grinding" all-nighters, and he was getting nowhere. His design goal was to produce lights that could be "subdivided;" that is, run in parallel circuits, and not just serial. The eventual solution was to use high resistance 100 ohm filaments instead of 5 ohm arc lights. But in 1879, no scientist on earth imagined that a high ohm light could exist. When Edison later claimed he had one, very few scientists believed him.

Edison had revealed the phonograph April 1878. He was proclaimed the scientist of the ages, he showed the phonograph in the offices of Scientific American, and at the White House... The reporters found out that he had patented 158 inventions, and by the fall of 1878 he was acclaimed the "Napoleon of Science" and "Wizard of Menlo Park." Arc lights and other experimental incandescent lights had been around for about 15 years, but they were not practical or economical compared to gaslight. There was no question that the phenomenon existed, and there should have been no question that if anyone could bring it to commercial reality, Edison could. But he hadn't done it yet, he was being lambasted in the press and the scientific establishment. He was busy doing science by press conference: "Popping into the newspaper pages like a vaudeville announcer going onstage to usher in each new act. Edison set off a scientific brouhaha of unprecedented proportions... William Siemens, who had worked on electric lighting for a decade, declared: 'Such startling announcements as these should be depreciated as being unworthy of science and mischievous to its true progress.'" (p. 129)

By and by, Edison got the thing to work, so he strung up several lights on poles around his laboratory. Ordinary people came every evening from miles around to marvel. The scientists stayed home. They knew the answer already, they didn't bother replicating, they did not even bother going over to have a look at a public demonstration. The drumbeat of daffy skeptical opposition continued. Professor Henry Morton, who knew Edison personally, lived nearby, and could have driven his buggy over any time, "felt compelled 'to protest in behalf of true science.' The results of Edison's experiments, he asserted, were 'a conspicuous failure, trumpeted as a wonderful success. A fraud upon the public.'"

The "prestigious" Professor Du Moncel said, "One must have lost all recollection of American hoaxes to accept such claims. The sorcerer of Menlo Park appears not to be acquainted with the subtleties of the electrical science. Mr. Edison takes us backwards." A letter in Scientific American attacked the light bulb and the newly improved dynamo, saying it would be "almost a public calamity if Mr. Edison should employ his great talent on such a puerility." Edwin Weston, a respected arc light specialist called Edison's claims, "so manifestly absurd as to indicate a positive want of knowledge of the electric circuit and the principles governing the construction and operation of electrical machines." (p. 162) The reaction in Europe was even more negative, and the "English Mechanic" journal declared "all anxiety concerning the Edison light may be put to one side. It is certainly not going to take the place of gas." (Anxiety?)

Finally, the light began to dawn (as it were). For one thing, people realized that Edison was, after all, a world famous inventor applying for a license to illuminate a neighborhood in New York, and he was getting huge amounts of carte blanche financing from the world's most canny, successful investors, including J. P. Morgan. People who had their head screwed on straight began to wonder whether J.P. knew something they did not know. People realized that J.P.'s men had good technical judgement — just as any sensible person in 1993 realizes that when MITI and Toyota start spending millions a year, it cannot be because they have accidentally mistaken 20C for 80C, or 0 watts for 100 watts.

Gradually, the "daffy skeptic brigades" swung into absurd denial, phase II. 'Okay,' they admitted, 'maybe these weird effects do exist after all, but we see now that they will never be of any use to anyone.' Here is a mind-boggling warning from the New York Graphic, in an article about the incandescent light: "When the phonograph was invented and the telephone was paraded before the admiring public, promises of magical results were lavishly made. How signally they have failed of perfection everyone now knows."

Three years after the telephone had been introduced, the daffy skeptics were busy writing it off as a useless toy! 'It may exist,' they said, 'but it is marginal, don't sell your telegraph stocks yet!' They knew that Bell had been wrapped up in litigation for years, and could not raise capital, and could not get to work, and they knew that he was fighting the biggest, most powerful vested interest in the world: Western Union. They knew that the development was tied up in knots by hostile opposition and severe technical challenges, but they figured three years should be enough to bring it to fruition no matter what the obstacles. It was all over, they said, the telephone was a failure. This is the same nonsense we hear today from some skeptics who expect people to conduct and publish clean, beautiful CF experiments even though they know that there is not a single dollar of money; that most universities strictly ban all work (even when the researchers offer to pay for the equipment); that workers face unending hostility and accusations of fraud; and that no journal editor will print even a short letter about CF. It's all over, he says, because the number of articles has declined.

Phase II denial always kicks in after a while. The establishment lambasted the Wrights for five years as bluffers, frauds, and liars, and never bothered to send a reporter to Dayton. The Scientific American and the New York Times never acknowledged the photographs and affidavits (just as today they refuse to acknowledge videos, data, and letters from MITI). Occasionally, a positive note would creep into S.A., but it would immediately be countered by another strong denial. The experts made up their minds instantly, the moment they heard the news in 1903, and that was that. Even after the 1908 demonstration flight at Ft. Myres, VA, one of the leading skeptics jumped up and said 'okay, it's real, but they will never be able to carry a passenger.' They already had carried a passenger, and they did it again a week later. History does not record what the skeptic said after that, but I am sure it was something like: 'okay, but it is unreliable, dangerous and expensive; it will never be practical.'

So, here we are 115 years after the incandescent light, 90 years after the airplane, and we have just entered "absurd denial phase II" with CF. Here is Dr. David Williams, Professor of Chemistry at University College London, lambasting cold fusion in Physics Today: "Indeed, it is important to say that there do seem to be some good measurements which indicate the possible occurrence of an interesting phenomenon. But what profit is there in such an inefficient, unreliable, dangerous and expensive energy storage method?"

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We are preparing the next issue of this newsletter.
Would you like to submit a contribution?
an announcement?

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Editorial Cartoon
adapted after Fusion
Facts (2/92), entitled
"DAVID &
GOLIATH
REVISITED"

"Cold-Fusion" DAVID
takes on "Hot-Fusion"
GOLIATH with greater
output and greater
"active"-time!!



COLD FUSION TIMES

*** The Short Life and Coming Hard Times of Bad Publishing ***

"Bad Science: The Short Life and Very Hard Times of Cold Fusion"
Random House, June, 1993, 461 pages, \$25.00
by Gary Taubes

A Critical Review by Eugene F. Mallove, Sc.D.

The promotional flyer, the very first page of the reviewer's "uncorrected proof" copy of *Bad Science*, shouts this small-minded book's preposterous bottom line: "In a fast-paced news-breaking account that reads like a novel, noted science journalist Gary Taubes brings to life one of the greatest scientific frauds of our times."

Remarkably, author Gary Taubes contends that cold fusion research is a wide-spread scientific fraud perpetrated by thousands of scientists, most of whom are obsessed with a quest for wealth and fame from a non-existent phenomenon. Because of his stubborn belief in this alleged "fraud", Taubes dismisses the continuing scientific efforts of thousands of cold fusion research scientists and engineers in over a dozen countries and at scores of institutions -- including the U.S. Electric Power Research Institute (EPRI), the Japanese Ministry of International Trade and Industry (MITI), and numerous corporations -- mostly Japanese, but a growing number of U.S. companies. That accusation of globe-girdling fraud is a pretty big tale to tell, especially when you are talking about the likes of NTT in Japan, one of the largest telecommunications giants in the world. In late 1992 NTT announced that one of its research groups had found a major piece evidence to support cold fusion -- helium-4 found by high-resolution mass-spectrometry.

Taubes plays the old "Big Lie" routine, this time masqueraded as science journalism. He says that the "few researchers" still working in the field aren't motivated to acknowledge negative results, because that would terminate their funding. Thus, he says, cold fusion promises to be "an endless loop," a fraudulent one at that.

Taubes is bold and blunt in his summation: "Within six months of the announcement of cold fusion [on March 23, 1989], its public life had deteriorated into a dismaying struggle against reality in which the believers explained the insipid state of their science with all manner of causes, none of which was as simple as the reality itself. Cold fusion -- as defined by Stanley Pons and Martin Fleischmann, or Steve Jones, or as modified by John Bockris or Edmund Storms or Carol Talcott, or Bob Huggins-Stanford, or whomever -- did not exist. It never had."

The *Bad Science* promotion says "no one was able to duplicate the findings" of Pons and Fleischmann, and that is exactly what Taubes says in his book. Mr. Taubes, a science journalist who claims at least superficially to be very thorough (he said he interviewed 260 people through November 1992), apparently forgot to check the scores of technical papers on cold fusion that have been published since September, 1989 in *Fusion Technology*, a journal of the American Nuclear Society. He also forgot to examine many other fine technical journals, including *Physics Letters A*, the *Japanese Journal of Applied Physics*, half-a dozen conference Proceedings, etc., etc. Just as *Bad Science* is about to go to press, *Physics Letters A* of May 3, 1993 has a peer-reviewed cold fusion calorimetry paper by Drs. Pons and Fleischmann.

The article describes convincing evidence of massive excess energy of non-chemical magnitude, including reproducible boiling conditions in cold fusion cells that generate several kilowatts of excess energy per cubic centimeter of palladium cathode. This work was presented openly by Drs. Pons and Fleischmann last October at the Third International Conference on Cold Fusion in Nagoya, Japan. Video tapes of the boiling cells were shown publicly. Mr. Taubes wasn't there, nor was he at the Second Annual International Conference on Cold Fusion in Como, Italy in 1991. It is highly unlikely that Mr. Taubes will attend the Fourth International Conference on Cold Fusion in December, 1993, which will be sponsored in Hawaii by EPRI, the research arm of the U.S. electric utility industry.

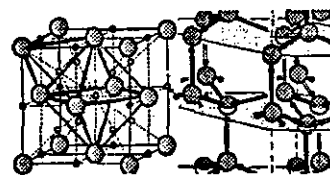
The promotional flyer of *Bad Science* says that Random House will send the author on a tour to include at least New York, Washington, D.C., Salt Lake City, and Los Angeles. The publisher plans to advertise the book, in *The New York Times* (*Science Times*). Strange, because in the November 17, 1992 issue of *Science Times* there appeared the revealing article by Andrew J. Pollack, "Cold Fusion, Derided in U.S., Is Hot in Japan," which elaborates the vitality of the field, while also quoting American negativists of Mr. Taubes's ilk. *Bad Science* is certainly part of that derision, as was Mr. Taubes's 1990 unsubstantiated and later disproved accusation in *Science* magazine that cold fusion cells were probably deliberately spiked with tritium at Texas A&M University.

Allegations of fraud in the adulteration of cold fusion cells with tritium are dredged up at length *again* in Taubes's sordid and tedious account in *Bad Science*. A high-risk game for Mr. Taubes, one would think, because tritium has since been found in numerous cold fusion experiments at dozens of laboratories ranging from the Bhabha Atomic Research Center to Los Alamos National Laboratory, to the U.S. Naval Research Laboratory. Some laboratories have achieved "completely reproducible" tritium generation.

*** A Simple Message ***

The message of *Bad Science* is as simple as it is pernicious: nuclear reactions at low energy (temperature) are impossible, therefore all the experimental results from cold fusion experiments are either: (A) The result of experimental incompetence or (B) Fraud. The part of the book in which we are told what happened after the Utah announcement of March 23, 1989 is called "Book II: A Collective Derangement of Minds." That is another theme of Taubes: people who came to believe that there was anything to the evidence for cold fusion were captured by a global scientific hallucination, they were drawn to the exciting hunt for excess heat and nuclear products and were ensnared in mass delusion. To Mr. Taubes, it was only the "smart" fellows at Caltech, MIT, Harwell Laboratory, and Yale, etc. who got it right, did "good" experiments, decided cold fusion was nonsense after they "failed" to confirm Pons and Fleischmann, and then left the field exhausted and disgusted.

Mr. Taubes tells the cold fusion saga in chronological order, but he mysteriously manages to arrive at page 400, still not out of cold fusion's first year! From page 400 to page 423 he covers another year to March 1991, by page 427 Taubes is in June 1992, and then he devotes only one more page in the "Epilogue" to everything thereafter. This is a measure of how Taubes views cold fusion -- he thinks it lost all credibility in 1989, but it continues to live on in some weird self-perpetuating, near-death state.



OP - ED REVIEW PAGE

Taubes calls cold fusion "a small, however perverse, subject" and yet he almost seems to regret that he had to relegate what would have been several hundred more pages of main text to 40 pages of detailed chapter notes. There Mr. Taubes places what he calls "second or even third level of information, speculation, perspective, and humor in which readers can delve if they choose." Taubes claims that the "social phenomenon" of cold fusion "peaked" in May, 1989 and has gone downhill since then. This proves that Taubes doesn't know what he is talking about. It is certainly his "wish" to see cold fusion dead or dying, but that is not reality. In this death wish he is not alone; he has many compatriots in the pack-journalism community, which is to say much of journalism. To suggest in 1993 that cold fusion is "asymptotically dying" requires enormous chutzpah or mental confusion, or both. On both accounts Mr. Taubes is well-endowed.

In his very first sentence Mr. Taubes makes two assertions and then spends the rest of his book egregiously violating his own precepts! This is a marvelous joke on Taubes, who is ever the glib dispenser of jibes at others, though it may not be too much longer before Taubes realizes that the joke is on him this time. He writes: "The cold fusion episode teaches two lessons that can be applied as meaningfully to journalism as to science:

1. Do your research, because nothing is as simple as it seems.
2. Make sure you've got the story right before you publish."

Taubes neither got the story of cold fusion right before he published, nor did he realize that cold fusion is not as "simple as it seems." He bought hook, line, and sinker the scientific establishment's incorrect, self-satisfied view of cold fusion as "pathological science." That anti-scientific view was unleashed by a relatively small band of like-minded media manipulators and negativists.

Taubes confidently explains it all. He says that many cold fusion scientists seem to have "renounced a life of science for one of faith." He says that an atmosphere of "blind faith" pervades cold fusion. But Taubes himself has a foolish "faith" that "all is known" about what goes on in hydrided metal lattices, a simplistic view that boggles the mind. He has this comment, "All those cold fusion proponents who were claiming that something magical happens within the lattice of the palladium were strenuously neglecting the fact that what happens in the lattice was already understood." The hydrided palladium lattice is nothing, if not incredibly complex. It has superconducting properties, unusual bonding characteristics, etc., which are by no means well understood. The nuclear effects that have now unambiguously emerged from cold fusion investigations, are an entirely new dimension of solid state physics.

Taubes has another peculiar "faith" -- faith that nuclear physicists know that there can't possibly be anything of a nuclear character brewing within the palladium lattice. He writes, "...the absence of neutrons strongly implied, if one had any faith in nuclear physics, that the tritium was not formed in the cell at all. Rather it must have come together, so to speak, elsewhere and entered the cell in a more circuitous manner." Mr. Taubes, means fraud -- deliberate spiking of cells at Texas A&M. What we have from Taubes is a shocking new operating condition in science:

continued on page 9

Well, back to Earth. I'd like to read you a letter which I sent to Vice-President Gore last week: it should have reached him by now:

Dear Mr Gore,

18 March 93

COLD FUSION. (?) I am happy to learn that you are being briefed on the above - perhaps misnamed - subject, as it is impossible to imagine anything of greater potential importance from both the economic and geopolitical points of view.

After initial scepticism, I have now seen so many positive reports from highly respected organizations (e.g. NTT - which is already marketing experimental kits in Japan' - ONR, US Army. Research Office, SRI, MIT) that there can be no further doubt that excess energy is being produced by some previously unknown process, not essentially nuclear. I am sure that your staff has already seen much of this material, and I also refer you to Representative Swett's statement in the Congressional Record for 16 February, 1993.

Whatever the source of the energy - which I am sure will be elucidated in the fairly near future - the sixtyfour trillion dollar question is: (1) is this merely a laboratory curiosity of no practical importance, or (2) can it be scaled up for industrial and perhaps even domestic use? If No (2) is correct, the consequences are immeasurable. it would mean essentially the end of the 'Fossil Fuel Age', and an era of cheap, clean power. The environmental benefits would be overwhelming; at the very least, concern with CO2 build-up and acid rain would vanish.

Clearly, no effort should be spared to resolve this matter speedily, by supporting scientists who are obtaining results (and, perhaps, discouraging those who have been obstructing them.) One witness you might call is my friend Dr George Keyworth II President Reagan's Science Advisor and an expert on fusion physics, who remarked in a recent letter to me: "the conventional path we've been pursuing is trying to build a bridge across the seas instead of inventing a boat." Perhaps 'Cold Fusion' may give us the lifeboats Spaceship Earth so badly needs!

Respectfully,
Arthur C Clarke

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2. 'Anomalous Nuclear Reactions in Condensed Matter.. A report on the Third International Meeting on Cold Fusion'.. Dr Iqbal Ahmad (US Army Research Office [AMC] Far East)
3. 'Summary of Third International Conference on Cold Fusion in Nagoya'. Peter L Hagelstein, Massachusetts Institute of Technology - Research Laboratory of Electronics
4. 'Nuclear Product Ratio for Glow Discharge in Deuterium'.. A B Karabut, Ya Kocherov and I. B Savvatimova. Physics Letters A 170 265 (1992)
5. 'Deuterated Metals Research at SRI International': 4 Mar 1993

converting harmful radioactive isotopes into nonradioactive elements at room temperature using properly constructed electron environments. Consequently, one of the troublesome aspects of fission energy might be eliminated.

The phenomenon will eventually be understood and useful devices will be constructed. While this process is evolving, the current absence of U.S. policy is inhibiting development of a vital legal and intellectual property infrastructure. At the present time, very few scientists in the U.S. have a competent understanding of the field. Because this field is so new and complex, training people will not be an easy or rapid task. In addition, the U.S. patent office is not issuing patents in the field while this limitation does not exist in other countries. As a result, control of the technology is slipping away on these two important fronts. A modest amount of support now can have a significant impact on reducing both problems without the need to shift major priorities before the potential of the phenomenon is better understood. I would like to recommend three actions.

First, a panel of open-minded technical experts be appointed to evaluate recent results.
Second, in the interim, it is prudent to provide some resources in view of the increasing activity of other countries. This money would extend the understanding of this complex field and help train a larger number of scientists in its subtleties. Efficient use of these limited resources could be achieved by supporting laboratories that have reported positive results in the past.
Third, encourage the Patent Office to start issuing patents in the field

Dr. Bogdan C. Maglich's lengthy testimony lashed out at the DOE's narrow-minded fusion energy program and defended his brand of aneutronic fusion energy. He defined "aneutronic fusion" as fusion which "burns nonradioactive fuels, produces 100 times less neutron radiation, 1000 times less radioactive waste and 2-3 times less heat pollution than the conventional fusion." Maglich's company, Advanced Physics Corporation, is the lead member of the USA-Russia Research Consortium, S.A.F.E. System for Aneutronic Fusion Energy. The consortium consists, he said, of 12 member organizations. Some of the American members of the Consortium are: the University of California at Irvine, Academic Lead Member, Advanced Reactor Group of Professor L. Lidsky, Nuclear Engineering Department of MIT; General Electric Corporate R&D Center, Schenectady, NY. The Russian membership consists of six major institutes involving 138 scientists. Its Lead Member is the nuclear weapons center Chelyabinsk-70, whose 42 scientists have decided to convert from military to peaceful research by developing aneutronic fusion technology.

Some excerpts from Dr. Maglich's testimony: "Large development programs acquire institutional and psychological momentum, and change much more slowly than our understanding of the facts. As a result, the DOE fusion energy program has taken no cognizance of modern scientific developments, nor has the DOE taken necessary steps to encourage the development of successor programs. Billions of dollars of taxpayers money would be saved and dozens of years of national economic development would be shortcut by making the national fusion program recognize that scientific tests within the past five years (1988-93) supported by theoretical calculations strongly point to the feasibility of making the fusion reactor COMPACT."

"Tests in two tokamaks carried out independently in 1988-89, show that ultra hot fuel does NOT behave 'anomalously' - which would require a gigantic reactor - but behaves 'classically' - which implies a SMALL reactor. Independently, measurement in the new type of small fusion reactor known as SELF-COLLIDER, published in Physical Review Letters of January 18, 1993 show how ultra hot ions of 10 billion degrees can be made STABLE in a miniature reactor whose volume is 100 cc. As a result of a series of experiments with compact models of aneutronic fusion reactors, our research consortium is now only 3 years away from proving once and for all that a nonradioactive-fuel, nonproliferating, aneutronic fusion energy reactor is feasible. Aneutronic fusion research involves a modest fixed cost. Because the devices are small, the cost of an experiment is in the \$3-5 million range, and the cost of an entire scientific test is in the \$30 million/3 years range. The system consists of two parts: injector and self-collider reactor. Proof of principle can be tested on a time scale of 3 years. If successful, this new fusion reactor could become commercially available 5-10 years thereafter, depending on plant size. DOE cannot afford not to pursue the new laboratory measurements coupled to the Air Force supported computer simulations indicating the feasibility of fusion reactors burning non-radioactive-fuels - (mixture of nonradioactive helium-3 and deuterium, as opposed to that of radioactive tritium and deuterium). For the past 30 years, the US DOE has systematically refused to fund any research on non-radioactive-fuel fusion as a matter of policy. Its support has been narrowed to one type of fuel and one type of reactor. This is in clear violation of the letter of the law and congressional intent as set forth in the Energy Policy Act of 1992 (Public Law 102-486 of 10/24/92). The first fusion goal mandated by the Law is 'a broad based fusion energy program.' The law also gives priority to environmentally clean energy technologies and to programs in which private industry participates 20% or more. The current DOE radioactive fuel program has reserved only \$500,000 out of the \$340 million, or 0.1%, of the Magnetic Fusion Energy Budget for alternate fusion energy option. Nonradioactive fuel fusion research would be a small part of this 0.1%."

"An institution can never be made to compete with itself - which would be the case if this program is administered by the present Office of Fusion Energy of DOE. To facilitate the development of such law-mandated, fusion ecotechnology effort, the Congress should direct the Department of Energy to split the present Office of Fusion Energy into: * OFFICE OF NONRADIOACTIVE-FUEL FUSION ENERGY; and * OFFICE OF RADIOACTIVE-FUEL FUSION ENERGY (ITER); and That separate budgets be sought for the two offices, with the goal that, over the next three years, at least 50% of the fusion budget be allocated for non-radioactive fuel fusion."

"It had become increasingly clear that the problems associated with tritium based fusion would be nearly insurmountable in a commercial device. Power plants burning deuterium-tritium fuel would necessarily be large, radioactive, highly toxic (beryllium), highly flammable (lithium), complex, and because of low power density, very expensive per unit of electricity generated. Tokamak could be made marginally economical only if it breeds and sells plutonium - which is incompatible with the US nonproliferation policy."

"In spite of these facts, the fusion program has changed little since the 1960's, even though many of the assumptions that appeared reasonable at the time have been found, in the course of 30 years, to be invalid. The goal became not to develop a commercially-viable, environmentally benign fusion-powered electric source, but rather to prove that the original decision to build a D-T burning tokamak reactor was the only possible decision."

So, with the testimony of Mills, Storms, and Maglich, the Subcommittee members received an eye-opening education on the possible future direction of the Fusion Energy program. There could well emerge from this hearing and others that will surely follow, much more open-mindedness in Congress being the ponderous creature it is, it may take some time, but given the spirit of the May 5th hearing, I would not even rule out limited funding in FY '94 for the non-tokamak concepts that were discussed - including cold fusion. It seems to me that the hot fusion program is in very big trouble at this hour. It may only manage to survive via its considerable political momentum, which is getting to be less and less as Congressmen increasingly examine the facts.

To summarize the significance of the May 5, 1993 Fusion Energy Hearing:

1. The hearing was the first time since the late April 1989 House Science Space and Technology Committee hearing that cold fusion had received an extensive public review before Congress. It was noteworthy that cold fusion and hot fusion were discussed at the same forum - a landmark event.
2. We now have "at least" three Congressmen who are solidly open-minded about cold fusion, as demonstrated by the May 5 lines of questioning: Congressman Swett of New Hampshire, Walker of Pennsylvania, and Fawell of Illinois. Dick Swett, as his post-hearing press release indicates, has become decidedly pro-active.
3. The top aides of the HSTT Committee appeared to be very interested in doing what they could in the future to help investigate cold fusion. This from conversations directly after the four hour hearings. There is considerable interest in helping Mills, in particular, get a working device evaluated at a national laboratory. There is further interest in having an eventual demonstration perhaps even in the Congressional offices!
4. The hot fusion people were not very well received. This was the same old stuff, promises, promises, promises, with not much new concrete to report - except that Princeton has 200 Curies of tritium on site and they are just dying to contaminate the hell out of the TFR tokamak when they start their D-T testing in September. Congressman Fawell and Walker gave the HF boys (and gals) a hard time. Fawell challenged them with the recent public words of Robert Hirsch. Hirsch thinks the D-T program is going nowhere. In these tight budget times I think Congress is going to scrutinize HF much more closely. If CF works out the way many expect it to in the coming year, I think there is a serious chance that Congress will reduce HF funding, and may completely kill it for FY '95.
5. The HF people acted as though CF did not exist. They made no acknowledgement of it as an issue, nor they asked about cold fusion. One wonders what they could or would have said. They have not been following the experiments, so presumably they would have had nothing to say other than "beware, it's pathological science."
7. There may be extremely promising fallout from these hearings, there may even be a shot at getting some cold fusion funding approved for FY '94. I think there will be a symbiotic effect between (A) the accepting atmosphere in Congress, (B) new experiments and scientific papers being and soon to be reported-- particularly all the light water work, and (C) increased media attention. The whole situation is unstable.

Arthur C Clarke, CBE

Dear Mitchell,

May thank for your letter of 30 March and enclosure.

I've read Cold Fusion (I love your article) with interest, and with it the book of last. Another copy arrived separately, and I've sent it to a distinguished person friend (the world's first professor of Invention!) who must believe in CF. I think it will shake him up.

Meanwhile, here's what say to me of the most important: because I ever saw (note the italics!) you are at liberty to quote from it, but only short extracts because I intend to revise it as soon as possible for general publication.

As you'll see from my schedule, I'll be a non-person for the next few years, but I'll still following the developments in the field with avid interest. Incidentally, my new novel The Mirror of the Mind "Cold Fusion" for general.

All good wishes,

ACF

Dr Arthur C Clarke, CBE

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COLD FUSION TIMES

CONTINUED FROM PAGE 8

The Short Life and Coming Hard Times of Bad Publishing

Beware of reporting results that in any way seem to grossly conflict with "accepted" wisdom -- especially if the results are of a sporadic character. Because if you do report such new findings, muck-raking "science journalists," egged on by the rigid establishment, may accuse you of fraud.

*** Character Assassination ***

Bad Science is replete with extended character assassinations. Stanley Pons bears the brunt of much of Taubes's fury, although Steven Jones of BYU is also mercilessly reviled. In fact, Taubes gives credence to the long suspicion by Fleischmann and Pons that Jones "borrowed" their ideas, contrary to the prevailing impression circulated by Frank Close and others that Fleischmann and Pons were energized and egged-on by their initial visit to the BYU-Jones laboratory. Martin Fleischmann is attacked less than Pons or Jones. He fares better, but not that much.

It is very difficult to distinguish fact from fiction in Taubes's account of the University of Utah-BYU interaction, but he does build a case that may be convincing to many people. He writes, "The evidence, oddly enough, for Pons and Fleischmann's version of the events is persuasive." He suggests that Jones, in effect, forced the early public disclosure by the University of Utah as he attempted to gain credit for his own meager neutron results and supposed origination of electrochemically induced fusion ideas. Taubes uses a lot of strong words and phrases for Jones in this connection: "the piracy scenario which Pons and Fleischmann later constructed," "...he [Pons] was being robbed," rumors of "scientific larceny." Taubes notes that Jones in his denials of misconduct, "...was the man who doth protest too much." No doubt, Steven Jones will find much to criticize and be outraged about. Dr. Jones apparently allowed Taubes to view his journals and notebooks, but Taubes turns these against him.

Taubes is brutal about cold fusion theorist MIT Professor Peter Hagelstein. He says Hagelstein "concocted" a flimsy, flawed theory to explain non-existent cold fusion. He cites an unnamed researcher who describes the "Hagelstein myth" as a "myth of scientific creativity warring in one person against morality." Taubes's style is to use the words of others to disparage those he faults. So he has MIT Professor Ronald Parker, then head of the Plasma Fusion Center say about Hagelstein, "Some people are geniuses if you just measure their IQ, but that doesn't necessarily make people creative, or give people the verve to discover or be curious. So it is only a necessary condition, being smart. It doesn't guarantee success." This is so ironic, because Parker was one of the most obtuse, uncreative scientists in cold fusion. Parker is a bureaucrat of the tokamak program, who defended his big project against new ideas with the intensity and ruthlessness one has come to expect from such people. We read of an anonymous physicist at MIT who says of Hagelstein's cold fusion papers that they reminded him of one of his students in an undergraduate course "who couldn't do physics to save her life." We also read that DOE's William Happer in the Office of Energy Research thought of Hagelstein's ideas as "crazy, absolutely crazy."

Of Texas A&M University cold fusion researcher Professor John Bockris Taubes says: somewhere along in his scientific career he "began propagating flamboyant errors in lieu of good science."

He cites physics Nobel laureate Norman Ramsey of Harvard as "the most credulous" of the DOE panel members who investigated cold fusion. He

says that while Ramsey "was off hiking," panel co-chairman Huizenga had spent the summer of 1989 poring over cold fusion reports. Ramsey, says Taubes, wanted to "go soft" on Pons and Fleischmann.

Taubes praises journalists who ignored cold fusion even in its early days, and he mocks those who have given it any credence. To Taubes, Jerry Bishop of The Wall Street Journal, who has written extensively about cold fusion, is merely a "gullible" journalist who is singularly responsible for the continuation of cold fusion research. Says Taubes, "With the help of The Wall Street Journal cold fusion began to play like a nuclear version of the emperor's new clothes."

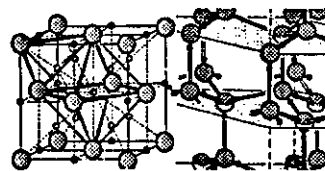
Bad Science is so permeated with offenses, it is difficult to catalog them. We find sick anecdotes, such as reference to the announcement of cold fusion falling on the sixth anniversary of Utah artificial heart patient Barney Clark's death. Taubes revels in scatological descriptions of all manner, such as the Caltech team's reference to Steve Jones's Mother Earth Soup electrochemical solution as "piss-green solution with gray slag at the bottom" or "brownish red shit with this much garbage on the bottom."

There is a recurring undercurrent of inappropriate anti-Utah and anti-religious sentiment throughout. For example, Taubes cites a literary and cultural critic, Paul Fussell's "...general principle that no high-class person can live in any place associated with religious prophecy or miracle, like Mecca, Bethlehem, Fatima, Lourdes, or Salt Lake City." He mocks the claims of cold fusion as "the discovery of salvation."

*** The "Good" Parts ***

Taubes certainly didn't learn very much about the science of cold fusion in his vaunted taped interviews. But his marvelous little recording machine has gloriously preserved the bigoted, pathetic remarks of the opponents of cold fusion research. Years before this sordid book came onto the scene, I predicted that these revelations would help document the shallowness of the so-called "scientists" who attacked cold fusion. It dredges up their nastiness.

DOE Cold Fusion Panel member Dr. William Happer told Taubes of an interview he had with a Scientific American reporter a few days after the Utah announcement: "The thing I didn't have the nerve to do was to say that just by looking at these guys on television, it was obvious they were incompetent boobs." Marvelous objectivity by a high-level DOE official! An apt description of how Dr. Happer handled himself in DOE's "investigation" of cold fusion is just that: incompetent boobyery. The cold fusion scientific community can rejoice that this high level boob has recently been given the boot at DOE. We learn from Taubes that Happer fancies himself a "tabletop experimentalist." Too bad he didn't take screwdriver in hand and perform cold fusion experiments himself. He was too busy jiggling wires and finding fault with everything he saw in his travels -- Mr. Know-it-All. There are today professors at small colleges who are miles ahead of Happer in their knowledge of experimental procedures in electrochemistry and calorimetry. These scientists consistently measure excess heat. Happer only gave other people heat. We get from Taubes that Happer said Bockris had "this collection of pathetic students and postdocs." What



9

a Wonder Man that Happer, he can tell a bunch of incompetents on first meeting!

My former MIT friend Richard Petrasso is unmasked in his hot-tempered glory. We learn many times from Taubes that Petrasso had a "distaste for glib bullshit," meaning the Pons and Fleischmann press conference. Petrasso's nearly uncontrollable rage against cold fusion is evidenced in many quotes in Bad Science. I didn't need those quotes, however; when I was at MIT I witnessed Petrasso's seething hatred of cold fusion on many occasions. Taubes recounts how Petrasso, in preparing his critique of Pons and Fleischmann's neutron evidence for Nature, had to be restrained by his colleagues from preserving highly emotional language.

We already knew his public arrogance, now we meet the privately arrogant Nathan Lewis of Caltech, who apparently initially told a colleague that Pons and Fleischmann's experiment "wasn't worth more than one day's effort." Taubes quotes Lewis's self-praise, "So I'm one of the people, where you say, 'If Nate says it's right, it's right.'" Great guy that, "Nate," the man whom Nature magazine has protected from damning technical criticism of his 1989 cold fusion paper by refusing to publish scientific correspondence from at least three other scientists. Taubes loves Nate: "...what Lewis published tended to be the last word." He is apparently fond of Lewis's colorful perception that cold fusion would be "an electrochemist's wet dream" -- IF it were true.

And we learn that Lewis's soul brother at MIT, then Professor (now Provost) Mark Wrighton, only "grudgingly decided" to attempt the Pons and Fleischmann experiment. Taubes loves Wrighton too: "And, like Nate Lewis who had studied under Wrighton, when Wrighton published, his papers were expected to be definitive." Well, the 1989 MIT Plasma Fusion Center paper on cold fusion calorimetry for which he is a co-author is anything but "definitive" -- to put it very charitably! The incorrect analysis and data massaging on which that paper was based is by now very well known.

Thanks to Taubes we now know that physicist Michael Salamon of the University of Utah had strong "scientific" motivations for assailing cold fusion research. Salamon muses that in the unlikely event there were really something to cold fusion, "It would be the cruelest, most unjust act of God to bestow Stan with this. There is no one less deserving."

*** Mocking Truth and Decency ***

Mr. Taubes review lacks common decency. Not even the dead are safe from his attacks. Taubes has the unmitigated gall to quote a former associate of Stanley Pons, Kevin Ashley, concerning the death of electrochemist Andrew Riley in the January 2, 1992 cold fusion experiment explosion at SRI: "The worst sin of this whole affair is [Riley's] death." Dr. Riley was extremely dedicated to the cause of cold fusion research, and like his colleague Dr. Michael McKubre of SRI, felt it to be a subject of extraordinary importance. One might

Continued next page

"A bill of rights is what the people are entitled to against every government on earth, general or particular; and what no just government should refuse, or rest on inference."

Thomas Jefferson.
Letter to James Madison (12/20/1787)

The Short Life and Coming Hard Times of Bad Publishing

say Andrew Riley died a hero in the quest for scientific understanding and truth. His memory did not need either Mr. Ashley's and certainly not Taubes's vile epitaph. But this type of gutter "journalism," is not surprising for Mr. Taubes. A fellow science journalist for a major metropolitan newspaper once warned the MIT News Office about Taubes: "Beware, he's a hatchet man." Indeed, he is. After Taubes's savaging of Nobel laureate Carlo Rubbia in his book, Nobel Dreams, one gets to know the style.

Taubes has a weird theory of science. He seems to suggest that mass delusion will feed cold fusion indefinitely. The madness will not stop. He paraphrases DOE panelist, Alan Bard, who says that "cold fusion research will never end abruptly, because the tail of the distribution simply trails off indefinitely, edging ever closer to zero, but never becoming exactly zero. The horizon will always serve up yet another researcher, an iota farther out on the tail, who will be capable of botching the experiment, either consciously or unconsciously, and willing to say publicly that he has confirmed." These will be "sloppy scientists," "bad scientists," "opportunists," and those "working far outside their fields." Taubes actually muses whether cold fusion "pathological science" might be like a "kind of infection like the measles, that attacks the scientific community periodically and has to run its predestined course."

As Taubes's book amply shows, a band of scientific establishment people in the U.S. went after cold fusion and thought they could kill it. They failed, of course, but not before they severely curtailed cold fusion research in this country. In 1993, cold fusion is gradually winning acceptance as new results, new experiments, and new insights are gained about one of nature's most startling and spectacular mysteries -- the appearance of nuclear reactions "in the cold." Truth will overwhelm the scientific community and authors like Taubes. It is absolutely inescapable.

The era of pack-mentality on cold fusion is coming to an end. Scientists and journalists are beginning to come to their senses, just when Taubes thought he could be shielded and cheered by a pack of sheep. No doubt many scientific bigots will praise Bad Science and some journalists may wish to bless Taubes's distorted account -- as did William Broad of The New York Times in 1991, when on page one of the Sunday Times he gave wide currency to the negative message of Frank Close's book Too Hot to Handle. But it will be like flies landing on sticky fly-paper. The negativists will cheer this book as they have each of the two preceding negative accounts. In the final act, the flies and flypaper will be trashed forever.

"Men it has been well said, think in herds; it will be seen that they go mad in herds, while they only recover their senses slowly, and one by one." Just as in this Taubes quote of author Charles Mackay, cold fusion is advancing in the scientific community one person at a time -- as the herd madness of the anti-cold fusion camp crumbles. People are realizing that they have been bamboozled by rampant distortion and Big Lies. Unfortunately for the publisher and author, there is no Bigger Lie than the title of this book. Taubes and others who write obscene, preposterous books like Bad Science, will be among the last to come to their senses, but come they will, dragged, kicking and screaming all the way.

FOR THE WORLD (and more information):

We want to thank the many readers, contributors, supporters, helpers, consultants, and friends who have helped bring this magazine to fruition.

We also want to acknowledge and thank Malcolm Partridge for the printing assistance, John J. Mara for adaptation of his unparalleled drawings of the aqueous phase, our friends at the Memory Location for their constant presence and knowledge, and Prof. Arthur C. Clarke for contributing beyond serendipity.



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COLD FUSION TIMES

The Top Ten Cold Fusion Experiments

as selected by Russ George, The Cold Fusion Newsletter

Fleischmann and Pons original palladium cathode platinum anode in heavy water first announced March 1989.

Variations on #1 with vigorous boiling as performed by F&P and demonstrated via television and video broadcast beginning in the fall of 1992.

Takahashi - Cold fusion using palladium foils in electrochemical cells at the University of Osaka 1991 to present.

Palladium electrochemistry as performed by the Electric Power Research Institute (EPRI) and the Stanford Research Institute (SRI) 1989 to present.

The replication of the Takahashi experiment with variations as performed by Dr. Edmund Storms at Los Alamos National Laboratory in 1992.

The light ('ordinary') water, nickel, and potassium carbonate experiment of Dr. Randall Mills of Lancaster, Pennsylvania.

The light water, nickel, potassium experiments of Bush and Eagleton at California Polytech at Pomona.

The NIT gas loading experiments of Dr. E. Yamaguchi in Japan.

The ion implantation experiments of Takahashi at the University of Osaka.

The gas plasma experiments of Kucherov et al at the Lutch Poldsk Institute in Moscow.

Palladium electrode produces massive Tritium and Helium

review by Russ George, The Cold Fusion Newsletter

"On an electrode producing massive quantities of tritium and helium." J. Electroanalytical Chemistry, 338, 1992, Chun-Ching, Dalibor Hodko, Minevski, Bockris at Texas A&M.

A palladium electrode produced a concentration of tritium in a 0.1 M LiOD solution around 103 times above background. Tritium production ceased after a few days but could be restarted by a small increase of the deuterium over potential. Correspondingly, 4He was found in 9-10 pieces of the Pd electrode at 2-100 times background. Addition of fresh D2O quenched the tritium production which began again spontaneously after 1-2 days. If the tritium had come from contamination 3He would have been found in the electrode; it was absent. Only one in four electrodes examined produced tritium and 4He. The surface of this electrode contained a copper mosaic structure not seen on inactive electrodes.

{ Ed. The Cold Fusion Newsletter, a source of good background information, contains further reviews & data }

New York City

On June 24, 1993 Cold Fusion is planned to be the Focus of a Gala NYCity Meeting, "The Manhattan Project, Part II: PEACE"

Cold fusion researchers, technologists, and supporters - including author and futurist Arthur C. Clarke - will speak in Manhattan at "The Manhattan Project, Part II: Peace". This gathering is being held to promote public understanding about the revolutionary energy-from-water discovery that so captivated the world in 1989, only to be temporarily submerged by heated and occasionally irrational debate. Now the work of thousands of researchers in over a dozen countries - including a strong effort in Japan - has "confirmed" the controversial discovery, although a generally agreed-upon theoretical explanation remains elusive. Thus, the theme of the meeting: "Cold Fusion, an Awakened State!"

The event is sponsored by Deak Sonotech, Inc., a New York-based corporation that is working on cold fusion power systems and other cutting-edge products in the field of ultrasonics. The evening's program will feature MIT Professors Peter L. Hagelstein and Keith H. Johnson. Also speaking will be physicist-artist Dr. David Deak, President of Deak Sonotech and engineer-author Dr. Eugene Mallove, a Vice President of the company. Arthur C. Clarke, many steps ahead of the rest of the world in technological forecasting, is scheduled to address the meeting via a VideoPhone-link from Sri Lanka.

WHAT'S HAPPENIN'

As we went to press with this issue, the following timely letter was posted, from Dr. Melvin Miles of the U.S. Department of the Navy, Naval Air Warfare Center, Weapons Division, China Lake, CA, who reportedly asked Jed Rothwell to post this letter to Professor Jones on the internet: (sci.physics.fusion).

"Professor Steven E. Jones of Brigham Young University has been aggressively attacking my scientific publications involving the Fleischmann-Pons effect (cold fusion) for almost two years and has recently accused me of rejecting heat/helium data points and fudging my results. I highly resent such unfounded attacks since they damage my professional integrity as a scientist as well as reflect on the quality of research conducted at Navy laboratories. Navy scientists are required to submit their work for review and clearance before any manuscript can be released for publication.

My problems with Professor Jones began on July 23, 1991 when I presented an invited seminar on my research at Brigham Young University. I soon learned that this event was more of an inquisition rather than a normal scientific seminar. Professor Jones and his group interrupted with so many pointed questions that it took me nearly three hours to complete my seminar. Although I was upset with this situation, I tried my best to be patient and to answer all their questions. It was obvious that the real intent of this group was to find possible errors with my experiments. This was verified a few weeks later when Professor Jones mailed out a critique of my work to other scientists involved with this research field throughout the world. This eventually led to my published response to questions raised by Professor Jones regarding my work (see reference 1). This entire episode was clearly outside the bounds of professional decency towards an invited seminar speaker.

Professor Jones has recently made false allegations regarding my work by his statements of "rejected data points", "shifts the window", "fudging or worse", etc.. (see reference 2). Although Professor Jones tries to use some unknown author for several negative statements, the implications regarding my experiments are very clear. These damaging statements by Professor Jones are outright lies since there are absolutely no rejected helium measurements in my recent publication (reference 3). Every data point involving heat/helium measurements is reported. This can be readily verified by Dr. Ben Bush who did the helium analysis at the University of Texas as well as by my detailed laboratory notebook for these experiments. Although I remain open-minded to possible scientific errors in our measurements, there are no rejected helium measurements in our experiments. Furthermore, there is no fudging of any measurements.

The two data points that I discussed with Professor Jones at Brigham Young University in 1991 are samples 12/17/90-A and 12/17/90-B. Both of these data points are present in my Table 1 (reference 3). Professor Jones should have known better than to raise this false accusation a second time since I had responded to this same question previously (see reference 1, page 79). This repeating of damaging false accusations makes me question the professional integrity of Professor Jones. Honesty and professional decency should be very important attributes for staff-members of a church-related institution.

Although the implications of fraud by Professor Jones is my most serious concern, his lengthy discussions of my papers (enclosure 2) involves numerous other examples where he has misrepresented my experiments. I will submit a detailed listing of these additional false statements made by Professor Jones in a future letter. Scientific discussions are important and necessary for this controversial research area. However, any scientist who

sets out to discredit the work of another must be certain that he has the facts straight. It does not benefit anyone to mix fact with fiction and then distribute this to other scientists. The best format for the discussion of my work by Professor Jones would be in a refereed scientific journal. This would help assure that the debate would be based on reported experimental facts rather than on misconceptions and false allegations. I hereby challenge Professor Jones to take his allegations regarding my work to a refereed scientific journal. If he cannot do this, he should at least get the facts straight regarding my work or remain silent.

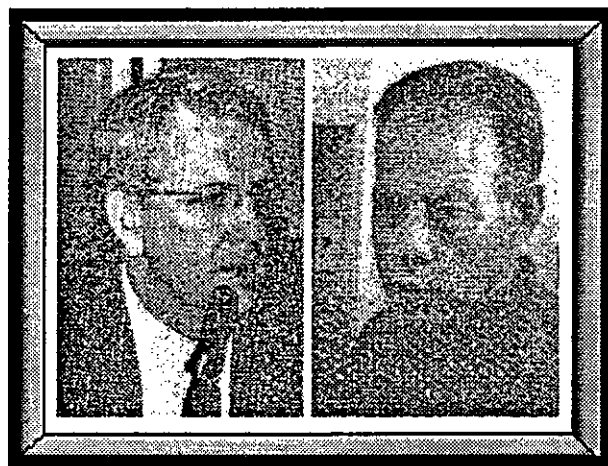
Please take note that my initial letter to Professor Jones was in response to reports of his constant criticism of my work on the computer network (E-mail). I merely asked him one simple question. I certainly did not ask him for detailed comments, criticisms, and false allegations concerning my publications.

Sincerely,

DR. MELVIN H. MILES
Ph.D. Physical Chemistry
NAWCWPNS Fellow

References

- (1) 21st Century Science and Technology, Vol. 6 No. 1, pp. 75-80 (1992).
- (2) Memorandum by Professor S.E. Jones, dated 27 May, 1993 and his E-mail discussion of my publications.
- (3) Journal of Electroanalytical Chemistry, Vol. 346, pp. 99-117 (1993).



Prof. Stephen Jones of Brigham Young University (left) and Dr. Melvin H. Miles of the Naval Air Warfare Center Weapons Division have continued their "discussions" for several months. These photos (taken by Jim Duree and Linda Sappington, respectively [21st Century Science and Technology (Spring '92 issue)]).

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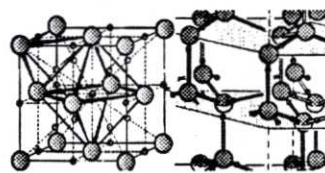
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HAS GONE BEFORE

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2001: THE COMING AGE OF HYDROGEN POWER

by Arthur C. Clarke, C.B.E.

"When .. World War II was still in progress, and I was working on Ground Controlled Approach Radar, which had the then enormous number of something like a thousand vacuum tubes in it, at least one of which would blow everyday. So it was impossible to believe, back in 1945, that TV relay stations could operate without a staff of engineers changing tubes and checking circuits. But of course, the transistor and the solid state revolution came along within a few years, and what I'd assumed would have to be done by large manned stations could be achieved by satellites the size of oil drums.

It is now beyond serious dispute that anomalous amounts of energy are being produced from hydrogen by some unknown reaction. The term 'cold fusion' - 'C/F' - has stuck because no one can think of anything better."

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